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S.B. PATIL COLLEGE OF SCIENCE & COMMERCE, RAVET

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UDISE NO: 27252001412

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Principle of Success

LIBRE OFFICE

From

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INTRODUCTION

- LibreOffice is a freely available, fully-featured office productivity suite.
- The LibreOffice suite is a collection of application programs for word processing, preparing spreadsheets, creating presentations, drawing diagrams, working with databases & composing mathematical formulae.
- Libre office is a open source software.
- It is platform independent software.
- LibreOffice has been translated (localized) into 40 languages.





APPLICATIONS OF LIBRE OFFICE

- Writer
- Calc
- Impress
- Draw
- base



INTRODUCTION TO BASE

- Libre office is a database software ,**similar to Microsoft Access, which provides a graphical interface for working with databases**
- Base is an open source database management system.
- Base is also called as relational database management software(RDBMS)
- Base is designed to allow users to easily create, access, modify, and view databases and their data like-tables, queries, forms, and reports.
- Base includes software wizards to assist users with various aspects of the program.
- **File extension for Base application is '.odb'.**



DBMS CONCEPT

- Database is a collection of related data items stored in a organized manner.
- Database consist of different objects like table, query, form or report.
- DBMS stands for Database Management Software.
- It is a software designed to define , manipulate, retrieve, and manage data(it means all kinds of facts, figures and details related to people , places, things or events) in database.
- It provides various function that allows entry, storage and retrieval of large quantities of information and provides a ways to manage that information.
- It also provide validation and manipulation of data.



HOW TO START LIBRE OFFICE BASE

- Click on start
- Select all programs
- Select libre office base

A database wizard will be started.

TO CREATE A NEW DATABASE

- 1] select “create new database “
- 2] click on next button
- 3] in save and proceed step click on “finish”
- 4] save as window will appear on the screen , select location to save the database.
- 5] type name for database in “file name” and click on save button.





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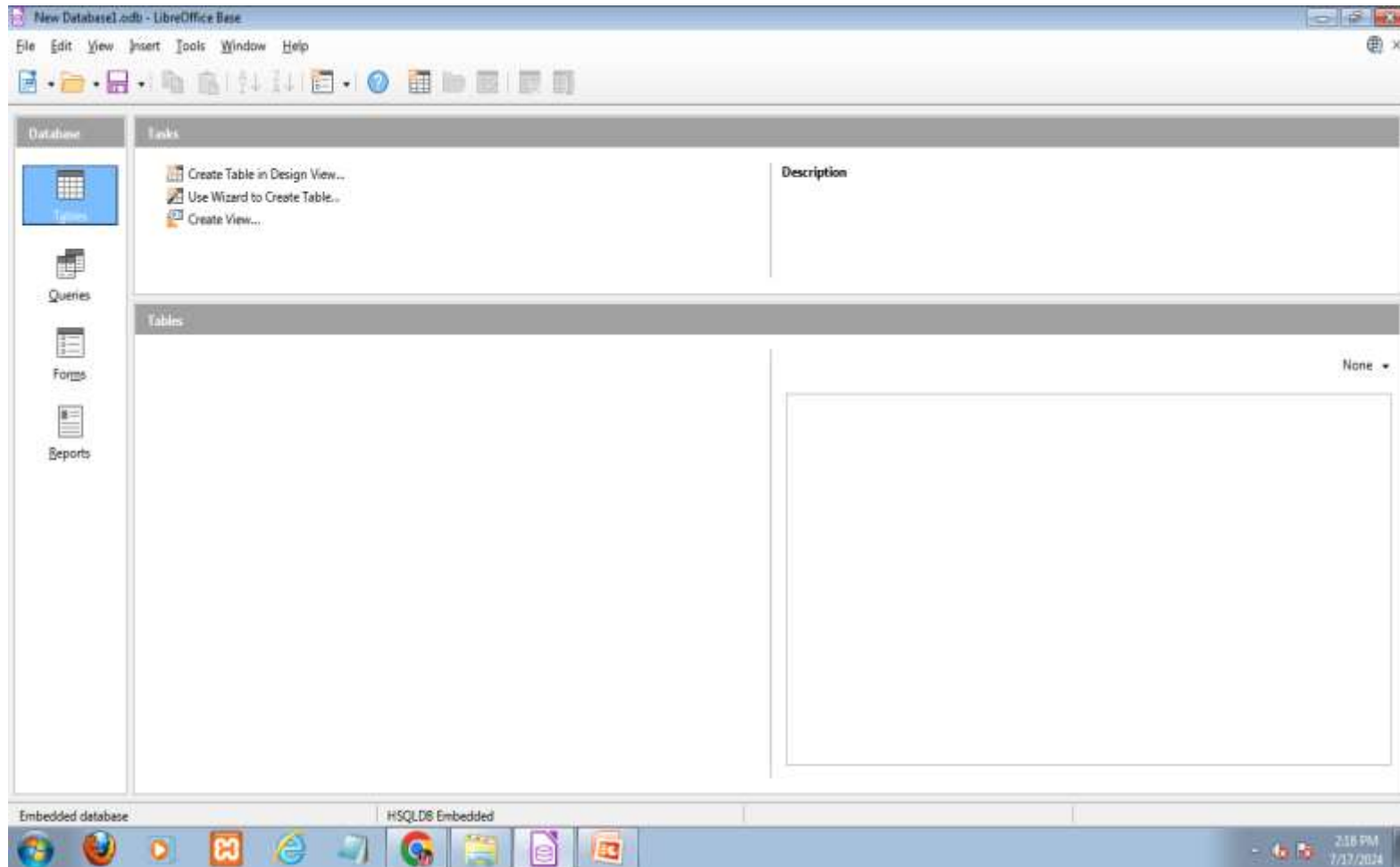
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LIBRE OFFICE BASE SCREEN



Mrs. Shailaja Pande

- Screen of base consist of following parts

- 1] Title Bar:- It is the topmost bar present on the screen of base. It display the icon of the application, name of the file and name of the application.
- 2] Menu bar :- It is present below the title bar, it display names of different menus.
- 3] Standard tool bar :- It consist of different icons which are used for standard operations.
- 4] Working area :- Below the standard tool bar called as working area. It divided into two pane left and right. left pane display name of the database objects like table, query, form, report. Right pane display activity related to that object.



Table

- 1] Table is a basic unit for storing data in database
- 2] table is a collection of rows and columns.
- 3] It is collection of related data.
- 4] The information stored under each column forms a row which is called as record/tuple.
- 5] column called as Fields

Datatype in base

Datatypes in base can be divided into three categories as alphanumeric, numeric, calender(date and time) and binary type.



QUERY

- LibreOffice Base **allow us to design, execute and store a query.**
- A query is a question asked within the database environment.
- Query display subset of data contained in various tables of database.
- Query is used to retrieve records from the tables.





FORM

- Form is used to collect data from the user
- Form object allows entering the data, editing or deleting existing data in the table.
- Form consist of format, style like radio button, list boxes that provides easy and systematic way to insert record in the table.



REPORT

- The presentation of information in an organised and readable format as per the users requirement called as report.
- Report is the representation of data in printed form .
- report are used to take decision .
- Reports are generated from the database tables or query.



INTRODUCTION TO DATA MODEL

- Data model defines how the logical structure of database is modeled.
- Data model defines how the data is connected to each other and how they are processed and stored inside the system.
- There are many types of data model such as
 - 1] Relational Data model
 - 2] Network data model
 - 3] hierarchical data model
 - 4] object – oriented data model
 - 5] entity-relationship data model



RELATIONAL DATA MODEL

- The most commonly used data model is relational data model.
- It is “ relational “ because the values within each tables are related to each other.
- A relational database refers to a database that stores data in a structured format , using rows and columns.
- In relational model, tables are called relations that stores data for different columns.



COMMONLY USED TERMINOLOGIES IN RELATIONAL DATA MODEL ARE :-

❖ Attribute

The columns of a relation called as attributes , it also called as fields

Attribute --- columns ----- fields

❖ Tuple

Each row of data in a table called as tuple.

Tuple ---- row ---cardinality

❖ Cardinality

The number of tuples in a relation called as cardinality.



○ Domain

A simple definition of a database domain is **the data type used by a column in a database**. This data type can be a built-in type (such as an integer or a string) or a custom type that defines constraints on the data.

Degree

The number of attributes in a relation is called the degree of relation.

Degree = Total no. of columns.



TYPES OF KEYS IN RELATIONAL DATABASE

○ Key

A key is a field, or combination of fields, in a database table used to retrieve and sort rows in the table based on certain requirements.

A key is also used when we want to establish relationships between the different columns and tables of a relational database.

Types of keys :-

- 1] Primary Key
- 2] Candidate key
- 3] Composite Key
- 4] Foreign key



Primary Key

- It is a unique key.
- The attributes used to uniquely identify the tuples in a relation is called primary key.
- It can identify only one tuple (a record) at a time.
- It has no duplicate values, it has unique values.
- It cannot be NULL.
- Eg:- STUDENT table Roll No. is a primary key

Candidate Key

A relation can have one or more attributes that takes unique values. any of these attributes can be used to identify the tuples in the relation. Such attributes called as candidate keys.



Composite Primary Key

A primary key having two or more attributes is called composite key. It is a combination of two or more columns.

Foreign Key

A foreign key is used to represent the relationship between two relations.

A FOREIGN KEY is a field (or collection of fields) in one table, that refers to the PRIMARY KEY in another table. The table with the foreign key is called the child table, and the table with the primary key is called the referenced or parent table.



TYPES OF RELATIONSHIP IN RELATIONAL DATABASE

- One – To – One

A *one-to-one* (1:1) relationship means that each record in Table A relates to one, and only one, record in Table B, and each record in Table B relates to one, and only one, record in Table A. Look at the following example of tables from a company's Employees database:





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○ One – To –Many

This is the most common relationship type

In this type, A row in table A can have one or many matching rows in table B, but a row in table B can have only one matching row in table A.

A one to many relationship can also be viewed as many to one relationship.



- Many – To -Many

In this relationship a row in table A can have many matching rows in a table B and vice versa.

Ex. a many-to-many relationship exists between customers and products: **customers can purchase various products, and products can be purchased by many customers.**

